Differential Geometry Excercise 3

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Question 1

Let X, Y be vector fields and let ω be a 1-form. This means that at each point $p \in M$:

$$X = \sum_i X^i \frac{\partial}{\partial x^i} \qquad Y = \sum_i Y^i \frac{\partial}{\partial x^i} \qquad \omega = \sum_i \omega_i dx^i \tag{1}$$

Where $x_1, ..., x_n$ is the local coordinates system at p. The exterior derivative of ω is then

$$d\omega = \sum_i d\omega_i \wedge dx^i = \sum_{i,j} \frac{\partial \omega_i}{\partial x^j} dx^j \wedge dx^i = \sum_{i,j} \frac{\partial \omega_i}{\partial x^j} \left(dx^j \otimes dx^i - dx^i \otimes dx^j \right) \tag{2}$$

where the last equality was derived from the definition of the wedge product. This form is a 2-form, i.e. a map $d\omega_p : T_pM \times T_pM \to \mathbb{R}$. Evaluating it for X and Y at point p gives

$$\begin{aligned} d\omega(X,Y)_p &= \sum_{i,j} \frac{\partial \omega_i}{\partial x^j} \left(X^j Y^i - X^i Y^j \right) = \\ &= \sum_{i,j} \left(\frac{\partial \omega_i}{\partial x^j} Y^i X^j - \frac{\partial \omega_i}{\partial x^j} X^i Y^j \right) \end{aligned} \tag{3}$$

On the other hand:

$$\begin{aligned} X(\omega(Y)) &= \sum_{i,j} \frac{\partial}{\partial x^j} (\omega_i Y^i) X^j = \sum_{i,j} \left(\frac{\partial \omega_i}{\partial x^j} Y^i X^j + \omega_i \frac{\partial Y^i}{\partial x^j} X^j \right) \\ Y(\omega(X)) &= \sum_{i,j} \frac{\partial}{\partial x^j} (\omega_i X^i) Y^j = \sum_{i,j} \left(\frac{\partial \omega_i}{\partial x^j} X^i Y^j + \omega_i \frac{\partial X^i}{\partial x^j} Y^j \right) \end{aligned} \tag{4}$$

And substracting them will give:

$$\begin{aligned} X(\omega(Y)) - Y(\omega(X)) &= \sum_{i,j} \left(\frac{\partial \omega_i}{\partial x^j} Y^i X^j + \omega_i \frac{\partial Y^i}{\partial x^j} X^j - \frac{\partial \omega_i}{\partial x^j} X^i Y^j - \omega_i \frac{\partial X^i}{\partial x^j} Y^j \right) = \\ &= \sum_{i,j} \left(\frac{\partial \omega_i}{\partial x^j} Y^i X^j - \frac{\partial \omega_i}{\partial x^j} X^i Y^j \right) + \sum_{i,j} \left(\omega_i \frac{\partial Y^i}{\partial x^j} X^j - \omega_i \frac{\partial X^i}{\partial x^j} Y^j \right) \end{aligned} \tag{5}$$

But as we saw before the first sum is $d\omega(X,Y)_p$ and it's easy to see that

$$\sum_{i,j} \left(\omega_i \frac{\partial Y^i}{\partial x^j} X^j - \omega_i \frac{\partial X^i}{\partial x^j} Y^j \right) = \sum_{i,j} \omega_i \left(\frac{\partial Y^i}{\partial x^j} X^j - \frac{\partial X^i}{\partial x^j} Y^j \right) = \omega([X,Y]) \tag{6}$$

And so:

$$\begin{aligned} X(\omega(Y)) - Y(\omega(X)) &= d\omega(X,Y)_p + \omega([X,Y]) \\ d\omega(X,Y)_p &= X(\omega(Y)) - Y(\omega(X)) - \omega([X,Y]) \end{aligned} \tag{7}$$

Question 2

Let X be a vector field on a manifold M, and let $(\varphi_t)_{t\in\mathbb{R}}$ be the associated one-parameter group of transformations. Let ω be a k-form, the Lie derivative is the k-form defined as

$$\mathcal{L}_X\omega := \frac{d}{dt}\varphi_t^*\omega|_{t=0} \tag{8}$$

Cartan's formula states that

$$\mathcal{L}_X\omega = i_X(d\omega) + d(i_X\omega) \tag{9}$$

Where $i_X\omega(-,\dots,-) := \omega(X,-,\dots,-)$ is the contraction of ω with X. Let us prove it first for 1-forms. Let ω be a 1-form, and let $p \in M$ and $x_1,\dots,x_n$ be a local coordinate system at p. Then ω and X can be written as:

$$\omega = \sum_i \omega_i dx^i \qquad X = \sum_i X^i \frac{\partial}{\partial x^i} \tag{10}$$

The derivative of ω is then:

$$d\omega = d\sum_i \omega_i dx^i = \sum_i d\omega^i \wedge dx^i + d^2x^i = \sum_i d\omega^i \wedge dx^i = \sum_{i,j} \frac{\partial \omega^i}{\partial x^j} dx^j \wedge dx^i \tag{11}$$

The contraction of ω with X, $i_X\omega$ can be written as

$$i_X\omega = \sum_i \omega_i X^i \tag{12}$$

And so the derivative of the contraction is

$$d(i_X\omega) = \sum_j \frac{\partial}{\partial x^j}\left(\sum_i \omega_i X^i\right) dx^j = \sum_{i,j}\left(\frac{\partial \omega_i}{\partial x^j}X^i + \frac{\partial X^i}{\partial x^j}\omega_i\right) dx^j \tag{13}$$

Finally, the contraction of $d\omega$ with X is defined using the definition of contraction:

$$i_X(\alpha\wedge\beta) = i_X(\alpha)\wedge\beta + (-1)^p\alpha\wedge i_X(\beta) \tag{14}$$

Where α is a p-form. This gives:

$$\begin{aligned} i_X(d\omega) &= \sum_{i,j}\frac{\partial \omega^i}{\partial x^j} i_X\left(dx^j\wedge dx^i\right) = \sum_{i,j}\frac{\partial \omega^i}{\partial x^j}\left(i_X(dx^j)\wedge dx^i - i_X(dx^i)\wedge dx^j\right) = \\ &= \sum_{i,j}\frac{\partial \omega^i}{\partial x^j}\left(X^j dx^i - X^i dx^j\right) = \sum_{i,j} X^j\left(\frac{\partial \omega^i}{\partial x^j} - \frac{\partial \omega^j}{\partial x^i}\right) dx^i \end{aligned} \tag{15}$$

Adding the last two expressions gives:

$$i_X(d\omega) + d(i_X\omega) = \sum_{i,j}\left(X^j\frac{\partial \omega^i}{\partial x^j} - X^j\frac{\partial \omega^j}{\partial x^i} + \frac{\partial \omega_j}{\partial x^i}X^j + \frac{\partial X^j}{\partial x^i}\omega_i\right) dx^i = \sum_{i,j}\left(X^j\frac{\partial \omega^i}{\partial x^j} + \frac{\partial X^j}{\partial x^i}\omega_i\right) dx^i \tag{16}$$

On the other hand, the pull-back of ω by a smooth function φ_t is:

$$\varphi_t^*\omega = \sum_i(\omega_i\circ\varphi_t)(\varphi^* dx^i) = \sum_i(\omega_i\circ\varphi_t)d(x^i\circ\varphi_t) = \sum_{i,j}(\omega_i\circ\varphi_t)\frac{\partial(x^i\circ\varphi_t)}{\partial x^j}dx^j \tag{17}$$

Differentiating with respect to t, we get:

$$\mathcal{L}_X\omega = \sum_{i,j}\left(X(\omega_i)\frac{\partial x^i}{\partial x^j} + \omega_i\frac{\partial X^i}{\partial x^j}\right) dx^j \tag{18}$$

Since

$$\frac{\partial x^i}{\partial x^j} = \delta_{ij} \tag{19}$$

the first term becomes a single sum:

$$\sum_{i,j} X(\omega_i)\frac{\partial x^i}{\partial x^j}dx^j = \sum_i X(\omega_i)dx^i = \sum_{i,j} X^j\frac{\partial \omega_i}{\partial x^j}dx^i \tag{20}$$

Renaming the indices in the second sum, we get the expression we got for the RHS, which means the formula is correct for 1-forms. Now let us see how it would act for sums and wedge proucts of forms. Since all the actions in the formula; pull-back, i_X, and exterior derivative, are all ditributive over addition and wedge produts, and since any k-form can be represented as a sum of wedge products of k 1-forms, and since the formula is correct for 1-forms, it is correct for all forms.

Question 3

part 1

Given the manifold $\mathbb{R}P^n$, let us show when it is orientable using the local diffeomorphism $\pi : S^n \to \mathbb{R}P^n$ that maps every vector in S^n to the line it spans in $\mathbb{R}P^n$. Also note $\pi(x) = \pi(-x)$. Remembering the S^n is orientable, there exists an everywhere non-vanishing n-form $\alpha \in \Omega^n(S^n)$. This chart is induced by $\mathbb{R}^{n+1}$. For every point where $x^1 \neq 0$:

$$\alpha = \frac{1}{x^1}dx^2 \wedge \cdots \wedge dx^{n+1} \tag{21}$$

Now let $\varphi : S^n \to S^n$ be the diffeomorphism $\varphi(x) = -x$. Then:

$$\varphi^*\alpha = \frac{1}{-x^1}d(-x^1) \wedge \cdots \wedge d(-x^n) = (-1)^{n+1}\alpha \tag{22}$$

Now let us assume that $\mathbb{R}P^n$ is orientable. Then by definition exists an everywhere non-vanishing n-form $\omega \in \Omega^n(\mathbb{R}P^n)$. In local coordinates $y = y_1, \ldots, y_n$, it can be written as:

$$\omega = g(y)dy^1 \wedge \cdots \wedge dy^n \tag{23}$$

For some function g. The pull-back of ω to S^n by π, $\pi^*\omega$ is an n-form and thus is related to α by some non-vanishing function h:

$$\pi^*\omega = h\alpha \tag{24}$$

Pulling back with φ, we get:

$$\varphi^*\pi^*\omega = (-1)^{n+1}h(-x)\alpha \tag{25}$$

But since $\pi(x) = \pi(-x)$, this should equal $\pi^*\omega$, and thus:

$$h(x) = (-1)^{n+1}h(-x) \tag{26}$$

This means that the non vanishing map $h(x)$ has parity $n+1$, but since it is not vanishing, it cannot be odd and so $n+1$ must be even, meaning that n is odd.

part 2

Let M be a smooth manifold. First let us assume it is orientable, and so there exists an everywhere non-vanishing n-form ω. Let $x_1, \ldots, x_n$ and $y_1, \ldots, y_n$ be two sets of local coordinates and let $p \in U_x \cap U_y$ be a point in their intersection. ω at point p can be written as:

$$\omega = f(x_1, \ldots, x_n)dx^1 \wedge \cdots \wedge dx^n \qquad \omega = g(y_1, \ldots, y_n)dy^1 \wedge \cdots \wedge dy^n \tag{27}$$

This gives the relation between the functions f and g:

$$f = \det\left(\frac{\partial y^i}{\partial x^j}\right) g \tag{28}$$

Using a coordinate transformations that flips the sign of f and g, we can find coordinate systems x' and y' for which f and g are always positive. In these coordinate systems:

$$f = \det\left(\frac{\partial y'^i}{\partial x'^j}\right) g \tag{29}$$

And since $f, g > 0$ then $\det\left(\frac{\partial y'^i}{\partial x'^j}\right) > 0$. On the other side, assuming that there is an atals such that for every two sets of coordinats $x_1, \ldots, x_n$ and $y_1, \ldots, y_n$ and $p \in U_x \cap U_y$ be a point in their intersection

$$\det\left(\frac{\partial y^i}{\partial x^j}\right) > 0 \tag{30}$$

Then one can define n-form at point p using both charts:

$$\omega = f(x_1, \ldots, x_n) dx^1 \wedge \cdots \wedge dx^n \qquad \omega = g(y_1, \ldots, y_n) dy^1 \wedge \cdots \wedge dy^n \tag{31}$$

And using relation 28, it can be seen that if ω is non-vanishing in U_x then it is not vanishing in g_x, which means that there is an everywhere non-vanishing n-form, and so M is orientable.

Question 5

Let N be a manifold, $f : N \to R$ a smooth function and $t \in R$ a regular value of f. And let $M = \{x \in N : f(x) \leq t\}$. Since t is a regular value, its preimage $f^{-1}(t)$ is a submanifold.
Then, for every $p \in f^{-1}(t)$ there is a chart (φ_i, U_i) such that $p \in U_i$ and for which $f(x) = t - x_n$ where x_n is one of the coordinates. Let then $\{(U_i, \varphi_i)\}$ be an atlas on N such that in every chart containing $f^{-1}(t)$ one of the coordintes is $x_n = t - f(x)$, and for all other charts within M, $x_n > 0$.
Those charts are a subset of the atlas such that $U_i \in M$ and $M = \cup_i U_i$ and for every $x \in M$, $x_n \geq 0$. Then, according to the definition, M is a manifold with a boundary $\partial M = f^{-1}(t)$.

Question 6

Let ω be a k-form over the n-torus $T^n = (S^1)^n$. Since $H^0 S^1 = H^1 S^1 \cong \mathbb{R}$. In other words, for S^1, the 1-form $d\theta$ is closed but not exact. There are n such 1-forms, meaning that for $k = 1$, $\dim H^1(T^n) = n = \binom{n}{1}$. For $k > 1$, there are there are $\binom{n}{k}$ wedge products of those 1-forms, which are linearly independent and in $H^k(T^n)$. Therefore

$$\dim H^k(T^n) \geq \binom{n}{k} \tag{32}$$

Question 7

Let ω be a compactly supported top form on $\mathbb{R}^n$ with $\int_{\mathbb{R}^n} \omega = 0$. From Poincare Lemma, ω is exact, so there is an $n-1$ form α such that $\omega = d\alpha$. Let us now find an α that is compactly supported. Let B be a ball such that $supp\ \omega \subseteq B$. According to Stokes theorem:

$$\int_{\mathbb{R}^n} \omega = \int_B \omega = \int_B d\alpha = \int_{\partial B} \alpha = 0 \tag{33}$$

So, from 17.24 in Lee's book, α is exact in $\mathbb{R}^n \backslash \bar{B}$. Let β be a $n-2$ form such that $\alpha = d\beta$ and ψ be a bump function that is zero inside B and 1 outside some ball B' that wraps B, then $\beta' = \alpha - \psi d\beta$ is supported only on B' and follows $\alpha = d\beta'$.

Question 8

Let $f : S^n \to S^n$ be a smooth, odd function. Let us show that it must have an odd degree, starting from $n = 1$. Let us look at the 1-form $d\theta$, its pull-back by f is

$$f^* d\theta = d(\theta \circ f) \tag{34}$$

And the integral is:

$$\int_{S_1} f^*\omega = \int_0^{2\pi} d(\theta \circ f) = f(2\pi) - f(0) \tag{35}$$

But, since f is odd:

$$f(-x) = -f(x) \tag{36}$$

And on the 1-sphere:

$$f(\theta + \pi) = f(\theta) + \pi \tag{37}$$

And specifically:

$$f(2\pi) = f(\pi) + \pi = f(0) + 2\pi \tag{38}$$

So we get:

$$\int_{S_1} f^*\omega = f(2\pi) - f(0) = 2\pi \tag{39}$$

While:

$$\int_{S_1} d\theta = 2\pi \tag{40}$$

So we get:

$$\deg f = \frac{\int_{S_1} f^*\omega}{\int_{S_1} d\theta} = \frac{2\pi}{2\pi} = 1 \tag{41}$$

Which is an odd degree. Now, let us assume that it is true for the n-sphere, and prove for $n + 1$. Let $x_1, \ldots, x_{n+2}$ be the natural coordinates of $\mathbb{R}^{n+2}$. Then everywhere $x^{n+2} \neq 0$ there is a local coordinates $x_1, \ldots, x_{n+1}$. Let us divide the sphere into three section:

$$\begin{gathered} S_+^{n+1} = \left\{x \in S^{n+1} : x^{n+2} > 0\right\} \qquad S_-^{n+1} = \left\{x \in S^{n+1} : x^{n+2} < 0\right\} \\ S_0^{n+1} = \left\{x \in S^{n+1} : x^{n+2} = 0\right\} \end{gathered} \tag{42}$$

Where S_+^{n+1} and S_-^{n+1} are bounded manifolds with the boundary S_0^{n+1} which is S^n. Let us now look at the integral of an exact (n+1)-form $\omega = f^* d\varphi$. Integrating:

$$\int_{S^{n+1}} \omega = \int_{S_+^{n+1}} \omega + \int_{S_-^{n+1}} \omega \tag{43}$$

Since f is odd, for every point $x_+ \in S_+^{n+1}$ there is a point $x_- \in S_i^{n+1}$ such that:

$$f(x_+) = (-1)^{n+1} f(x_-) \tag{44}$$

And so, using Stoke's theorem, the integral is

$$\int_{S^{n+1}} \omega = \begin{cases} 0 & nodd \\ 2\int_{S^n} f^*\varphi & neven \end{cases} \tag{45}$$

Then:

$$\int_{S^{n+1}} f^* d\varphi = 2\int_{S^n} f^*\varphi = 2deg(f)\int_{S^n} \varphi = deg(f)\int_{S^{n+1}} d\varphi \tag{46}$$

Which means the degree in f is the same for S_{n+1} as for S_n.

Question 9

Let $f : S^n \to \mathbb{R}^n$ be a smooth, odd map. Let us assume that for every $x \in S^n$, $f(x) \neq 0$. The normalized version has its image in S^{n-1}:

$$F = \frac{f}{|f|} : S^n \to S^{n-1} \tag{47}$$

And is odd as well

$$F(-x) = -F(x) \tag{48}$$

The upper hemisphere S^n_+ as it is described at the previous question, is a compact, oriented, smooth n-manifold with a connected boundary $\partial S^n_+ \cong S^{n-1}$. Restricting F to the boundary of upper hemisphere, it is then a map from S^{n-1} to itself that extends smoothly to S^n_+, then by the degree theorem its degree is zero. However, based on question 8, its degree must be odd, which is a contradication and therefore the assumption that $f(x) \neq 0 \forall x \in S^n$ is false.